



Beat gestures made by human-like avatars affect speech perception

Matteo Maran¹, Renske Rötjes¹, Anna R. E. Schreurs¹, Hans Rutger Bosker¹

¹Donders Centre for Cognition, Nijmegen, The Netherlands

matteo.maran@donders.ru.nl, renskerotjes10@gmail.com, anna.schreurs@ru.nl,
hansrutger.bosker@donders.ru.nl

Abstract

In face-to-face communication, several visual cues support speech perception. Even the timing of simple up-and-down flicks of the hand, called beat gestures, can convey word stress, changing what individuals hear (e.g., *CONtent* vs. *conTENT*). While beat gestures have been traditionally investigated in human-to-human communications, nowadays individuals increasingly interact with computer-controlled avatars (e.g., virtual assistants). The present study tested whether beat gestures produced by an avatar affect word stress perception, similarly to human gestures. Furthermore, this study tested whether a minimal visual cue such as a 2D moving disc can also affect speech perception. Beat gestures made by the avatar significantly affected speech perception, albeit slightly less than human-made gestures. The disc condition did not affect speech perception. The present work lays the foundation for the application of (beat) gesturing avatars, which could be used to boost speech intelligibility.

Index Terms: speech perception, audiovisual integration, avatar, beat gesture, prosody

1. Introduction

In face-to-face communication, speech perception relies on both the auditory information and the visual cues produced by the speaker [1]. For instance, seeing lip movements can change what individuals hear (McGurk effect [2]) and meaningful hand gestures support speech intelligibility [3]. Even simple up-and-down flicks of the hand, called “beat gestures”, affect speech perception. Beat gestures are the most frequently produced gestures in communication [4] and increase the perceived prominence of the linguistic unit they co-occur with [5]. For example, beat gestures increase the perceived sentential prominence of the word they co-occur with [6], a property commonly exploited in public speaking [7]. Notably, recent work shows that beat gestures affect speech perception even at the lexical level, biasing the perceived prominence of syllables within words, ultimately affecting the perceived lexical stress (e.g., hearing the noun *CONtent* vs. the adjective *conTENT*) [8]. In particular, a beat gesture falling on the first syllable of a disyllabic word (e.g., *content*) biases participants to hear the word *CONtent* (Strong-Weak, SW, stress pattern), while a beat gesture falling on the second syllable biases towards the word *conTENT* (Weak-Strong, WS, stress pattern). This effect, termed “manual McGurk” effect, has been robustly observed in both in-lab [8], [9] and online [10], [11] studies, across different

response modalities [10], and in two different languages (i.e., Dutch in the previously cited studies, Spanish in [12]).

The present pre-registered work tests if beat gestures affect speech perception also beyond human-to-human interaction. Advances in human-computer interaction and digital technology are leading to an increasing number of instances in which individuals do not interact with humans, but with virtual animated conversational agents (henceforth, “avatars”). For example, avatars can be employed as virtual salespeople in the field of online commerce [13], as virtual teachers in education [14] and second-language learning [15], or as virtual assistants in remote healthcare [16], [17]. Avatars’ beat gestures might increase speech intelligibility, facilitating comprehension in individuals with temporary or chronic hearing difficulties (e.g., due to aging, cochlear implants, etc.) or in non-native speakers. The bias in lexical stress perception induced by beat gestures could be of particular use in languages that have a large number of minimal stress pairs (i.e., pairs of words that differ only by lexical stress, such as *CONtent* and *conTENT*).

Previous work suggests that gestures affect the perceived human-likeness of avatars [18]. Beat gestures produced by an avatar were also shown to provide emphasis at the sentential level on the word they accompany [19]. However, whether avatars’ beat gestures affect lexical stress perception remains an open question, especially in light of the conflicting findings of a preliminary report¹. In that study, an avatar’s beat gestures affected speech perception in simulated cochlear implant hearing conditions when words were produced with an ambiguous (e.g., *content*, half-way between *CONtent* and *conTENT*) or infrequent (i.e., 12% probability of occurrence in the tested language [20]) stress pattern. Surprisingly, the avatar’s beat gestures failed to reliably affect non-vocoded speech perception when the stress cues were ambiguous, despite the fact that in this experimental condition visual cues could serve as a source of disambiguation. Thus, previous work suggests that beat gestures made by avatars might facilitate speech perception in individuals with hearing difficulties, while leaving open the question of whether they can also be relevant for normal hearing conditions. Notably, the visual cues provided by the avatar employed in that previous study might not have been very natural, especially in terms of kinematics and size. Moreover, the avatar lacked any lip movements, possibly limiting the extent to which it was considered the source of the produced speech. The present study tests whether beat gestures made by a more sophisticated avatar can affect speech perception, and whether they do so to the same extent as human-made gestures.

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A second open research question concerns whether some degree of human-likeness is essential for artificial visual cues to influence stress perception. Previous work showed that an isolated pointing hand moving up-and-down does not affect lexical stress perception [21]. Additionally, a study showed that a minimal visual cue such as a disc, mimicking beat gestures in timing and 2D trajectory, elicited de-activation of brain regions involved in processing beat gesture and speech synchrony [22]. It is likely that, already at early stages of development [23], individuals fine-tune audiovisual integration in communication to speech and visual cues produced by humans (i.e., the interlocutor), but not by non-human entities. The present study begins to explore this research question by testing whether a minimal moving visual cue, such as a disc having the same trajectory as a human beat gesture (henceforth “beat gesture” for readability, but at the theoretical level we do not consider this visual cue a beat gesture), might affect speech perception.

Our pre-registered hypotheses are the following. First, we expect that the timing of beat gestures made by a human-like avatar will affect lexical stress perception, possibly to a similar or reduced extent compared to human-made gestures. Second, we expect that a moving disc will not affect lexical stress perception, since it lacks human-likeness and agency.

2. Methods

The preregistration of this study can be found at <https://osf.io/746dm/> and deviations are explicitly mentioned in the present paper. Data, materials and code can be found at <https://doi.org/10.34973/3k98-1q60>.

2.1. Participants

The pre-registered sample size of 50 participants (39 females, 11 males; mean age = 21.86, range = 17-58) was reached after 64 participants were recruited using the Radboud Research participation system (SONA) and the researchers’ personal network, as 14 participants met the pre-registered exclusion criteria. All the participants were native speakers of Dutch, with normal or corrected-to-normal vision, who did not take part in previous studies testing beat gesture and speech integration conducted at Radboud University. Participants gave informed consent, as approved by the Ethics Committee of the Social Sciences department of Radboud University (project code: ECSW-LT-2024-10-16-89669). Participants recruited via SONA received course credit for participating.

2.2. Materials and experimental design

2.2.1. Materials

Participants watched videos (50 fps) with a human, an avatar or a disc (i.e., the three levels of the factor *Appearance*, Figure 1) while a Dutch single word out of four minimal pairs (*Content* | *conTENT*, “content” | “satisfied”, *SERvisch* | *serVIES*, “Serbian” | “tableware”, *VOORnaam* | *voorNAAM*, “first name” | “respectable”, *VOORruit* | *voorUIT*, “windshield” | “forward”) was produced. A beat gesture could be absent, or fall on either the first (BeatOn1) or second (BeatOn2) syllable of the produced word. The videos of the human condition were taken from the publicly available materials of a previous manual McGurk study [9]. In these videos the lip movements and beat gestures are coherent (i.e., BeatOn1 and lips producing a SW word, BeatOn2 and lips producing a WS word), but lip movements do not affect lexical stress perception when beat gestures are present [9]. The word could be produced with clear

SW (e.g., *CONtent*), clear WS (e.g., *conTENT*), or ambiguous (e.g., *content*) stress cues. The acoustically ambiguous condition was sampled from the 7-step phonetic stress continua in [9], involving gradual interpolation of the F0 contours from the clear SW and WS extremes (duration and intensity were set to constant stress-ambiguous values). Specifically, they correspond to the audio step 3 for all the minimal pairs, apart from *SERvisch* | *serVIES* (audio step 4), based on perceptual data [9] demonstrating close to 50% SW responses as well as good susceptibility to effects of gestural alignment.

Videos of the avatar condition were generated using Wonder Studio’s AI-powered motion transfer and animation tool (<https://wonderdynamics.com/>). Videos of the human condition were used as input, and the human actor was replaced via AI with the human-looking avatar “The professor”, while retaining the audio. Importantly, this avatar closely mimics the human movement both in terms of gestures and lip movements. Videos for the disc condition were manually created in a four-step procedure. First, individual frames were extracted from the human videos using ffmpeg (<http://www.ffmpeg.org>). Second, in each frame a full circle, similar in color to the human actor’s skin color, was added over the position of the hand, based on a pre-defined landmark on the index finger. Third, a monochrome background, similar in color to the human video, was added behind the full circle. Finally, using ffmpeg the frames were combined together in a video and paired with the audio taken from the initial human video. For all the levels of *Appearance*, catch trials (see Section 2.2.2) videos were created adding a white cross at the center of the video.



Figure 1: Example frames extracted from corresponding human, avatar, and disc videos. See examples in the online repository for the full videos.

2.2.2. Experimental design and procedure

The experimental design consists of the factors *Appearance* and *Beat* (BeatOn1, BeatOn2). The number of trials per condition is adapted from the so-called manual McGurk “mini-test” [10]. For each level of *Appearance*, the experiment included 20 experimental trials (5 per minimal pair) with a word produced with ambiguous stress cues and a beat gesture falling on the first syllable (BeatOn1), and 20 experimental trials (5 per minimal pair) in which the same acoustic token was paired with a beat gesture falling on the second syllable (BeatOn2). We also presented 32 trials in which a word was produced with clear stress cues (4 SW and 4 WS for each minimal pair) without a beat gesture in each appearance condition for perceptual anchoring. Finally, for each appearance condition we included 8 catch trials in which a word was produced with clear stress cues (1 SW and 1 WS for each minimal pair) without a beat gesture and in which a white cross appeared at the center of the screen. In these catch trials, participants were asked to indicate that a white cross appeared, thereby motivating them to keep their eyes open throughout the experiment.

The experiment was hosted on the online platform Gorilla [24]. After providing demographic information, participants performed a headphone screening test based on dichotic pitch [25], designed to exclude participants not using headphones. After a short practice (8 total trials, with all the three levels of *Appearance* shown, as well as catch trials) which could be repeated until participants were sure that they understood the instructions, participants performed the main task. In the main task, participants were asked to indicate whether they heard a SW or WS word. Each trial started with two members of a minimal pair presented in written form to the participant: the SW word (e.g., *CONtent*) on the left, and the WS word on the right (e.g., *conTENT*). After a fixation cross lasting for 500 milliseconds (ms), a video was shown to the participants. After the video, three response buttons were presented: the SW member on the left, the WS member on the right, and *IK ZAG EEN WIT KRUIS!* (“I SAW A WHITE CROSS!”; to respond to catch trials) below in the center. After clicking on one of the options or 4 seconds had passed, the next trial started after a 500 ms blank screen. A total of 240 trials (approximately ~25 min) was presented in a randomized order. Every 48 trials, participants could take a break. After the main task, a debriefing questionnaire was presented.

2.2.3. Pre-registered statistical analysis

Perceived stress was analyzed using a generalized linear mixed model (GLMM), using the R package “lme4” [26]. Perceived stress was coded as follows: SW = 1, WS = 0. The analysis focused on trials with ambiguous acoustic information (i.e., experimental trials with a beat). The fixed effect structure included the factors *Beat* (BeatOn1, BeatOn2), treatment-coded with BeatOn1 as reference level, and *Appearance* (Human, Avatar, Disc), treatment-coded with Human as reference level. The random effect structure was determined using the “buildmer” R package [27] and included random intercepts by *Item* (i.e., the minimal pair) and by *Participant*, together with random slopes for *Beat* by *Participant*. *P*-values were obtained using ANOVA Type II Wald χ^2 tests.

2.2.4. Exploratory analyses

Given the intermixed design it is possible to guess the disc-gesture relationship. Therefore, non pre-registered exploratory analyses tested whether an effect of *Beat* in the disc condition could be observed in participants who guessed what the disc represented. Based on their response to the debriefing question “What do you think the dot/disc represented?”, 35 participants were considered to have guessed (henceforth, “guessing participants”) since their responses included one or more of the following words: *hand* (“hand”), *arm* (“arm”), *klemtoon* or *nadruk* (“emphasis”), *cadens* (“cadence”), *toonhoogte* (“pitch”), and *intonatie* (“intonation”).

Two additional GLMER analyses were implemented: the first one including *Disc Guessing* as a between-subject factor, treatment-coded with “Naïve” as reference level, and the second one mirroring the pre-registered GLMER but analyzing data from the guessing participants only. The random-effect structures were determined using “buildmer” [27].

3. Results

3.1. Pre-registered analysis

We observed a significant main effect of *Beat* ($\chi^2 = 61.166, p < .001$) and a *Beat* $\times$ *Appearance* interaction ($\chi^2 = 47.263, p < .001$).

No significant main effect of *Appearance* was observed ($\chi^2 = 0.757, p = .685$). These results suggest that the timing of beat gestures affected the perceived stress, but differently in human, avatar, and disc conditions (Figure 2). The model estimates revealed a significant *Beat* effect for human ($\beta = -1.092, SE = 0.120, z = -9.112, p < .001$), a significant difference between human and disc for the BeatOn1 condition ($\beta = -0.423, SE = 0.102, z = -4.133, p < .001$), but not between human and avatar ($\beta = -0.106, SE = 0.103, z = -1.023, p = .306$). Finally, the effect of *Beat* was smaller in both the disc ($\beta = 0.968, SE = 0.145, z = 6.692, p < .001$) and avatar ($\beta = 0.295, SE = 0.145, z = 2.039, p = .041$) conditions compared to the human condition. Non-pre-registered post-hoc comparisons (Bonferroni-corrected), motivated by the *Beat* $\times$ *Appearance* interaction, showed that the effect of *Beat* was significant in the human and avatar conditions (both $p < .001$), but not in the disc condition ($p = 1$). No significant difference was found for the BeatOn1 condition between human and avatar ($p = 1$), while they both significantly differed from the disc ($p < .001$ and $p = .016$, respectively). No significant difference was found for the BeatOn2 condition between human and avatar ($p = .554$), while they both significantly differed from the disc (human: $p < .001$; avatar: $p = .004$).

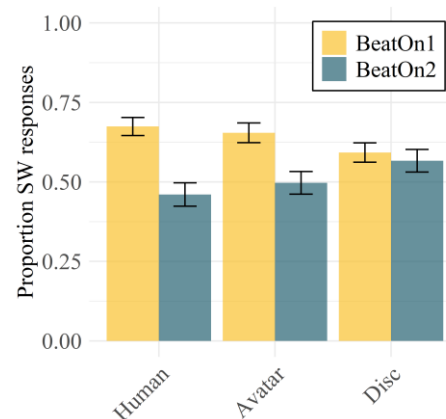


Figure 2: For each level of *Appearance*, the manual McGurk effect consists in the difference between BeatOn1 (left) and BeatOn2 (right). The y-axis indicates the mean proportion of SW responses. Error bars indicate the standard error of the mean.

3.2. Exploratory analyses

The exploratory analyses including the between-subject factor *Disc Guessing* revealed a main effect of *Beat* ($\chi^2 = 61.945, p < .001$) and a significant *Beat* $\times$ *Appearance* interaction ($\chi^2 = 46.915, p < .001$), replicating the pre-registered analyses. Crucially, no significant *Beat* $\times$ *Appearance* $\times$ *Disc Guessing* ($\chi^2 = 1.693, p = .429$) was observed (Figure 3).

In “guessing participants”, the disc condition resulted in a numerically larger proportion of SW responses for BeatOn1 compared to BeatOn2 (Figure 3). This small numerical difference is in the direction of the manual McGurk effect [8]. However, analyzing only data from these participants qualitatively replicated the results obtained with the full sample (*Beat*: $\chi^2 = 46.393, p < .001$; *Appearance*: $\chi^2 = 2.232, p = .328$; *Beat* $\times$ *Appearance*: $\chi^2 = 25.875, p < .001$). Bonferroni-corrected post-hoc comparisons, motivated by the significant *Beat* $\times$ *Appearance* interaction, showed a significant effect of

Beat in both the human and avatar conditions (both $p < .001$), but not in the disc condition ($p = 1$). While for the BeatOn2 condition both human and avatar significantly differed from the disc ($p < .001$ and $p = .012$, respectively), no such a difference was observed for BeatOn1 ($p = .163$ and $p = .278$, respectively), possibly due to a SW bias for the disc condition.

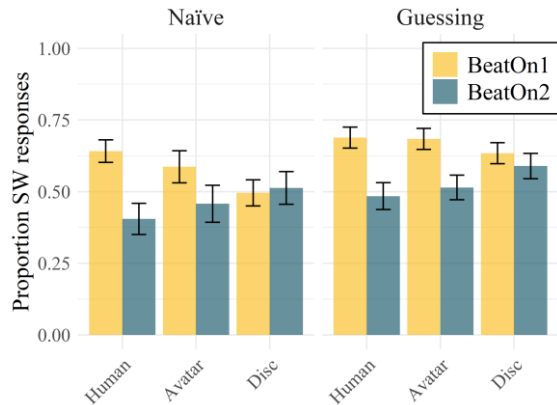


Figure 3: *Perceived stress data in Naïve and Guessing participants.*

4. Discussion

The present study tested whether the timing of beat gestures made by an avatar can bias lexical stress perception, as opposed to the movement of a simple disc. The results supported our pre-registered hypotheses. First, the timing of an avatar's beat gestures biased lexical stress perception, albeit slightly less than human beat gestures. Second, despite having the same 2D trajectory and temporal relationship with speech as a beat gesture, a moving disc did not bias lexical stress perception.

The results from the human condition replicate previous online manual McGurk studies. Notably, the present study employed an adapted version of the so-called manual McGurk “mini-test” [10], which tests gesture and speech integration in approximately 10 minutes. The results demonstrate how the “mini-test” can be used not only in isolation as a short test of human gesture-speech integration, but also in combination with other experimental factors (e.g., *Appearance*) to test relevant hypotheses in the field of psycholinguistics, while keeping the experimental sessions to an acceptable duration.

Our study demonstrates that the timing of beat gestures made by human-looking avatars can bias lexical stress perception. Accordingly, the present results might be integrated in modern applications of avatars, in particular by ensuring that the timing of beat gestures produced by virtual conversational agents aligns well with the prosody of their speech. The temporal alignment of gestures and prosody might be particularly important to boost speech intelligibility when virtual conversational agents interact with individuals with hearing difficulties (e.g., elderly population [28], or cochlear-implant users¹). Further applications can be found in the domain of virtual (second) language acquisition [15], where avatars' beat gestures might support the decoding of lexical stress and facilitate the acquisition of a correct pronunciation [29].

The significant manual McGurk effect observed in the avatar condition suggests that previous work¹ might have underestimated beat gesture-speech integration effects with virtual agents. It is possible that smooth and natural gesture

trajectories, such as the ones made by the avatar here employed, might increase audiovisual integration effects. Furthermore, the present avatar's lip movements might have reinforced the perception that the speech was indeed produced by the avatar. Parametric manipulations of these factors in future studies might reveal which features of the avatar's communicative visual cues affect the size of audiovisual integration effects.

Notably, the manual McGurk effect was smaller for the avatar than the human condition (BeatOn1 – BeatOn2: 0.16 vs. 0.21). This reduction might be due to differences in the appearance of the avatar, such as its larger hands and its cartoonish style, or in its body proportions. Another possible explanation, worth exploring in future studies, relates to the facial expression and eye gaze of the avatar (Figure 1). The avatar's videos were generated based on human videos in which the actor had a neutral face expression and fixed eye gaze (i.e., minimizing minimal eye movements). While our avatar was not perceived as strongly realistic or human-like (mean realism: 0.37, measured from 0 to 1 in the debriefing questionnaire), it is possible that such a neutral expression, when mimicked by the avatar, might have resulted in a feeling of uneasiness, similar to the “uncanny valley” [30] effect. Alternatively, the reduced effect might also stem from the fact that the avatar and the human condition shared the same voice, and therefore participants might have perceived the avatar as an artificial condition, based on the human video, with less sense of agency.

As predicted, a disc moving with the same 2D trajectory of a beat gesture did not affect lexical stress perception, not even in “guessing” participants. This might suggest that not all visual cues have affordances that allow them to affect communication (see also [21]). Through exposure, the human cognitive system might develop to selectively allow visual signals coming from a potential speaker, but not from non-human entities, to affect speech perception. Accordingly, temporal synchrony between speech and a visual signal might be necessary [31], but not sufficient, for audiovisual integration in communication, with the identity of the visual signal itself (i.e., potential speaker vs. impossible speaker) playing an important role. This hypothesis, possibly grounded in dissociations previously reported at the neural level [22], awaits further testing. It is important to note that the trajectory of the disc mimicked human beat gestures' trajectories only in two dimensions, and not in depth. An alternative hypothesis for future studies is that 3D movement, which characterizes the perception of biological motion [32], is a key component underlying beat gesture-speech integration. On a final note, in the disc condition a bias towards perceiving a SW stress pattern emerged (Figures 2 and 3). As SW is the most frequent (88%) Dutch disyllabic stress pattern [20], this finding suggests that when the audiovisual signal is ambiguous (i.e., ambiguous speech and no human-like visual cue), internal heuristics based on frequency might drive speech perception.

5. Conclusions

The timing of beat gestures made by a virtual avatar successfully biased lexical stress perception. Future applications of avatar technology might exploit the alignment of beat gestures and prosody to increase speech intelligibility. Conversely, a moving disc failed to affect speech perception, possibly due to the lack of affordances that make listeners consider it as the source of speech. The reduced influence of non-human minimal visual cues on speech perception might stem from selective tuning of audiovisual integration in communication to signals made by human-like agents.

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